NOVARTIS DISCOVERATHON 2026 · CHALLENGE 2 ·

Predicting hERG Channel Blockade

A Trustworthy Cardiotoxicity-Risk Model from Molecular Structure

Detailed Technical Report

Team: Fifth Paradigm · Institute: Indian Institute Technology Hyderabad · Date: 17th August 2026

Contents

Abstract

We built a machine-learning system that predicts hERG (KCNH2) channel blockade, a blocker probability and a pIC50 potency directly from a molecule's SMILES string. We assembled a 16,650-compound hERG bioactivity dataset from ChEMBL and BindingDB under a single set of quality gates, evaluated four independent model architectures on a leakage-controlled scaffold-split held-out set, and combined the two strongest and most complementary into a final wired model, the **MasterModel**.

On the held-out 20% test set the MasterModel reaches ROC-AUC ≈ 0.83, accuracy 0.75 and F1 0.74, beating every single model and every alternative blend. Crucially, our emphasis was **trustworthiness rather than a maximal leaderboard number**: a scaffold split that measures generalization to new chemistry, a confidence tier on every prediction (validated: high-confidence ROC-AUC 0.89 vs 0.71), a fully reproducible pipeline (regenerates its submission to 1e-4), and an honest external stress-test against 957 clinically-annotated drugs, where performance is deliberately bounded (ROC-AUC ≈ 0.61) for reasons we characterize precisely rather than hide.

1. Introduction

Drug-induced blockade of the hERG potassium channel (encoded by KCNH2) reduces the rapid delayed-rectifier current IKr, prolonging the cardiac QT interval and predisposing to the potentially fatal arrhythmia torsades de pointes [1]. hERG liability is a leading cause of late-stage drug attrition and market withdrawal, and its early assessment is mandated by the ICH S7B and E14 regulatory frameworks [2]. Reliable in-silico prediction of hERG blockade therefore lets teams' triage and re-design compounds before costly synthesis and electrophysiology.

Challenge 2 asks teams to assemble their own hERG dataset from public sources, train reproducible models, and evaluate them on an independent 20% held-out set using Accuracy, Precision, Recall, F1, ROC-AUC and PR-AUC. This report documents our data curation, splitting strategy, cross-validation methodology, the four model architectures, the final blended model and its weight-optimization, a full metric comparison, an honest external clinical validation, and the advantages, limitations and integrity of the work.

2. Data curation (training data)

We built one gated, from-scratch extraction that produces a single combined table carrying **both** the binary blocker label and the continuous pIC50 for every compound, from ChEMBL 37 [3] and BindingDB [4], under one consistent set of quality gates and one standardization. This replaced separate binary/regression pipelines whose mismatched provenance had left some compounds labelled but without potency (and vice-versa).

2.1 Sources and quality gates

Both databases are treated as first-class and extracted from scratch, and a row failing any gate is excluded (never silently re-admitted). Gates applied:

- **ChEMBL (target CHEMBL240):** assay-to-target confidence score ≥ 7; human target only; clean data-validity flag; potential-duplicate cleared; hERG mutant-assays removed (wild-type liability only).
- **BindingDB (hERG):** only parseable, curated activity values with usable units retained.

  Compound provenance was tracked after de-duplication: 14,110 compounds were found only in ChEMBL, 2,121 only in BindingDB, and 419 were present in both databases.

2.2 Activity definition and labelling

A hERG blocker is defined at a 10 µM threshold, i.e. pIC50 ≥ 5.0. Each measured row is labelled by endpoint with relation-aware handling: exact potencies (IC50/Ki/pIC50, relation '='/'~') give a direct label; inequality-censored values (e.g. IC50 > 10 µM → confident non-blocker) are retained as coarse labels; percentage-inhibition screens (≥ 50% at 10 µM ± 2 µM → blocker) contribute functional evidence; and assays describing channel activation are labelled non-blocker and mechanism-tagged. Genuinely ambiguous rows are excluded rather than guessed.

2.3 Standardization and per-compound aggregation

Structures are standardized with RDKit [5] (Cleanup → largest-fragment parent → uncharge) and re-keyed on the InChIKey, collapsing salts, charge states and cross-source duplicates. Rows are then grouped by InChIKey: pIC50 is the median of exact potencies, and the label follows a strongest-evidence precedence (exact potency → % inhibition → censored). Two quality flags are kept as columns without dropping compounds high_discordance (SD of exact pIC50 > 1.0 log unit) and label_conflict (rows disagree).

2.4 Final composition

Property	Value
Unique standardized compounds	16,650
hERG blockers / non-blockers	8,129 / 8,521 (48.8% positive)
Compounds with an exact pIC50 (regression)	10,316
Label source: exact / % inhibition / censored / other	10,316 / 2,544 / 3,787 / 3
Sources: ChEMBL / BindingDB / both	14,110 / 2,121 / 419

Table 1. Curated combined dataset composition (target CHEMBL240).

3. Data splitting and held-out design

3.1 Scaffold split, not random

We reserved 20% of compounds as an internal test set using a Bemis–Murcko scaffold split [6] via StratifiedGroupKFold (seed 42): compounds are grouped by their core scaffold and stratified by the blocker label, so the held-out set is class-balanced and, crucially, **scaffold-disjoint; zero scaffold overlap with training**.

This matters because chemical datasets contain families of analogs sharing a scaffold. A random split scatters near-identical analogs across train and test, letting a model appear accurate by near-memorization and inflating the reported metric by a well-documented margin [7]. A scaffold split forces entire scaffolds into train or test, so the held-out set contains chemotypes the model has never seen, it measures generalization to new chemistry, which is what actually happens in deployment. The reported number is lower but honest.

3.2 The split is representative

A harder split is only fair if it is not also a shifted one. It is not: the held-out set matches the training set on the target and the chemistry. The class balance is preserved (48.8% blockers in both), the pIC50 distributions are nearly identical (means 5.51 vs 5.50), and molecular-weight distributions overlap (means 435 vs 442 Da). The difficulty comes purely from unseen scaffolds, not a distribution shift.

Classification data: held-out preserves both class balance and chemistry

16,650 molecules split 13,319 train / 3,331 held-out (80/20, scaffold split, zero overlap); 48% blockers in both.

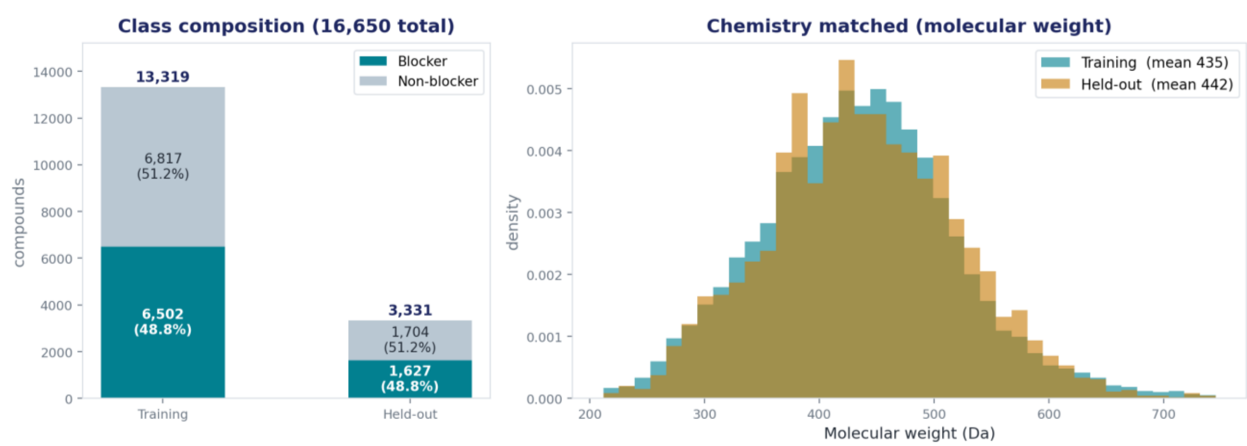


Figure 1. Classification data — training (13,319) vs held-out (3,331): class composition and molecular-weight distributions are matched.

Regression target (pIC50): held-out is scaffold-disjoint yet distribution-matched

10,316 molecules carry an exact pIC50 — Training 8,258 · Held-out 2,058. Same shape, means 5.51 vs 5.50.

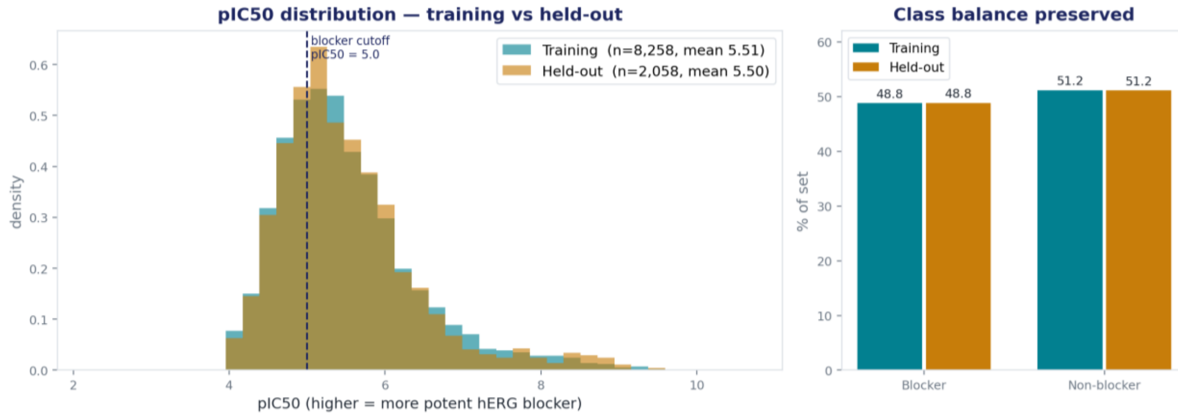


Figure 2. Regression target — pIC50 distributions of training vs held-out are nearly identical (means 5.51 vs 5.50).

Set	Compounds	Blocker %	With pIC50	Scaffolds
Training	13,319	48.82%	8,258	—
Held-out	**3,331**	**48.84%**	**2,058**	**1,586**
Overlap	—	—	—	0 (scaffold-disjoint)

Table 2. Held-out split (seed 42) distribution-matched yet scaffold-disjoint.

4. Cross-validation and why it matters

All model selection, calibration, threshold tuning and blend-weight fitting were performed with nested, scaffold-aware cross-validation on the **training pool only** (the 80% remaining after the held-out set was carved off). The untouched held-out set was scored exactly once, at the very end.

4.1 Nested, scaffold-grouped CV

The ML-Ensemble pipeline uses nested cross-validation with StratifiedGroupKFold (Murcko-scaffold groups) in both the outer and inner loops. The inner loop selects models and hyper-parameters; the outer loop provides an unbiased robustness estimate. Grouping by scaffold inside CV prevents the same analog leakage that a random split would cause the estimate reflects generalization, not memorization.

4.2 Out-of-fold predictions enable leakage-safe blending

Each base model emits out-of-fold (OOF) predictions on the training pool: every training compound is predicted by models that never saw it. These OOF vectors are what the blend weights are optimized against (Section 6), so the ensemble is fit without ever touching the held-out set, the reported held-out number is therefore not tuned. We additionally ran y-scrambling controls to confirm the pipeline learns real structure–activity signal rather than noise.

4.3 Why it matters

- Honest model selection: the best model is chosen on CV/OOF data, never on the test set.

- Leakage-safe stacking: OOF predictions let us fit blend weights without contaminating the test estimate.
- Robustness: nested CV shows the result is stable across folds, not a lucky single split.

5. Model architectures

We trained four architectures that represent a molecule in fundamentally different ways; a numeric feature table, a molecular graph, and a text sequence. This representational diversity is deliberate: models that see a molecule differently make different errors, which is what makes their combination effective (Section 6).

5.1 ML Ensemble (fingerprint / descriptor, gradient-boosted trees)

Each molecule is encoded as a 2048-bit Morgan/ECFP fingerprint (radius 2) [8] plus 24 RDKit 2-D physicochemical descriptors (MolLogP, TPSA, aromatic-ring counts, partial charges, fraction Csp^3, …). A model zoo; regularized logistic regression, random forest [9], extra-trees, XGBoost [10], LightGBM [11] and CatBoost [12] is evaluated in the inner CV loop; the top-3 by mean inner-fold ROC-AUC are combined by AUC-weighted soft voting, and probabilities are calibrated. It is the strongest single classifier (held-out ROC-AUC 0.823) and provides the deployed pIC50 regressor. It also emits the applicability-domain, boundary-instability and confidence-tier signals (Section 6.3).

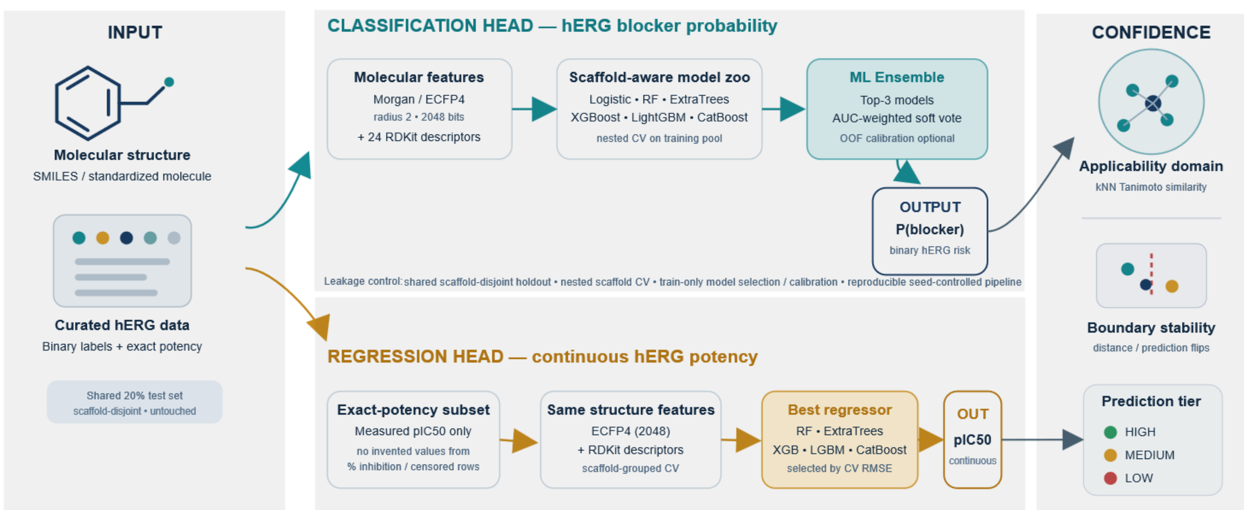


Figure 3. ML Ensemble: molecule → fingerprint + descriptors → gradient-boosted tree ensemble → calibrated probability.

5.2 GNN (graph neural network)

An AttentiveFP message-passing graph neural network [13] operates directly on the molecular graph (atoms as nodes, bonds as edges), learning connectivity-aware representations via graph attention. Held-out ROC-AUC 0.791. Evaluated as a base learner but not part of the final model (Section 6).

5.3 ChemBERTa as a Frozen-Embedding Featurizer

A ChemBERTa transformer [14] pretrained on large unlabeled SMILES corpora is used as a frozen featurizer; its embeddings are fused with ECFP4 and physicochemical descriptors and fed to a lightweight head. Held-out ROC-AUC 0.772. Exploration only.

5.4 FT-ChemBERTa (fine-tuned transformer)

The same ChemBERTa is instead fine-tuned end-to-end on our hERG data (the 'pure' checkpoint, no descriptor fusion). It reads a molecule as a token sequence, the way a language model reads text, bringing broad chemical knowledge from pretraining. Fine-tuning lifts it from ~0.77 to ~0.80 held-out ROC-AUC, and its errors are decorrelated from the tree model making it the ideal blend partner and the second component of the final model.

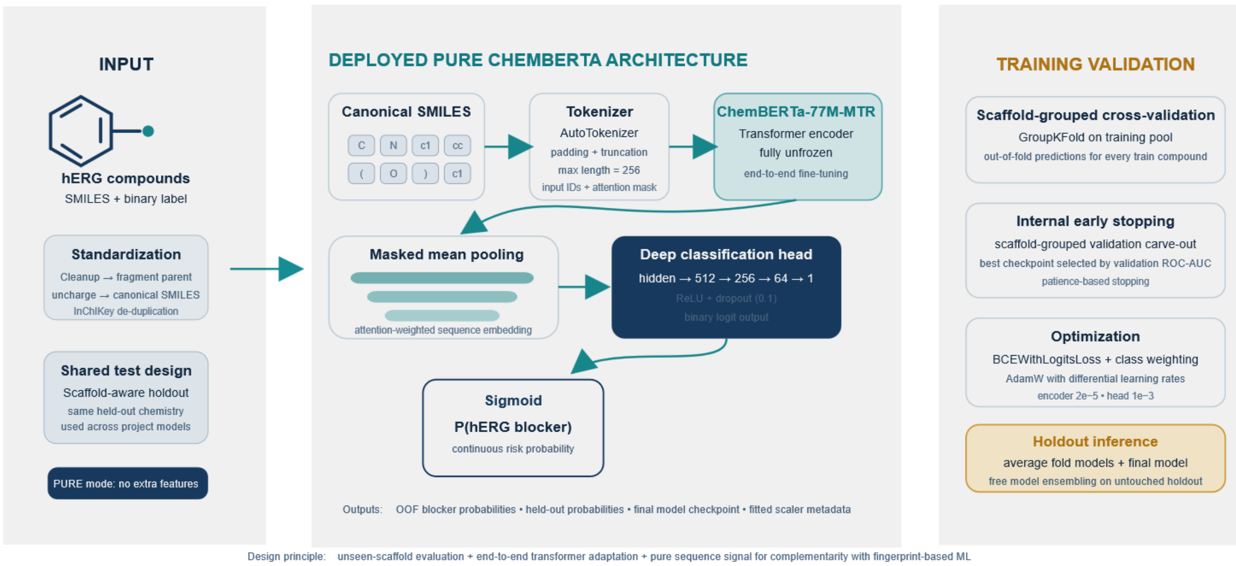


Figure 4. FT-ChemBERTa: SMILES sequence → tokens → fine-tuned Transformer encoder → probability.

6. Final model: the MasterModel and its blend

6.1 Architecture

The deployed model routes each target through whichever component won its own evaluation. For classification it is a weighted probability blend of the ML Ensemble and the fine-tuned pure ChemBERTa; for regression it is the ML-Ensemble regressor alone (every blend that touched the regressor worsened held-out RMSE/R^2, because the strong ML regressor lacks OOF predictions and averaging only dilutes it). No GNN is used in the final model. One SMILES enters; a blocker probability, a predicted pIC50, and a confidence tier come out.

6.2 The blend and its weight optimization

Given the base models' out-of-fold probability vectors Pi on the training pool, a convex weighted blend is

$$P_blend(w) = \Sigma_i w_i \cdot P_i, with w_i \geq 0 and \Sigma_i w_i = 1$$

The weights are chosen to maximize the leakage-safe out-of-fold ROC-AUC:

$$w* = argmax_w ROC - AUC(y_o of, P_b lend(w)) \text{ subject to the simple x constraints above.}$$

This objective is optimized by a simplex/grid search over the weight simplex, evaluated over every pairwise and three-way combination of the decorrelated base models entirely on OOF data, never on the held-out set. The winning combination and weights were the ML Ensemble and FT-ChemBERTa with

$$P(blocker) = 0.770 \cdot P(MLEnsemble) + 0.230 \cdot P(FT - ChemBERTa)$$

A logistic-regression stacking meta-learner, $P_b lend = \sigma\big(\beta 0 + \Sigma\beta i \cdot logit(Pi)\big)$,was also evaluated; the simple convex blend was selected for the deployed model as it was best (or tied-best) on OOF ROC-AUC while being simpler and more robust. The final weights were then applied once to the held-out set.

6.3 Confidence layer

Every prediction carries an applicability-domain (AD) flag the mean Tanimoto similarity of the compound's k = 5 nearest training neighbours against a training-derived threshold (≈ 0.494) [15], a boundary-instability flag (distance of the probability from the 0.5 decision boundary), and a combined HIGH / MEDIUM / LOW confidence tier (HIGH = in-domain and boundary-stable). On the held-out set the tiers are HIGH 1,916 / MEDIUM 382 / LOW 1,033.

7. Results: metric comparison across models

All numbers below are on the held-out 20% scaffold-split test set (N = 3,331), scored once. The four base models serve as the baselines; the MasterModel is best on every ranking metric.

Model	Acc	Prec	Recall	F1	ROC-AUC	PR-AUC
ML Ensemble (baseline)	0.746	0.744	0.731	0.738	0.823	0.821
GNN (baseline)	0.728	0.721	0.722	0.722	0.791	0.781
ChemBERTa featurizer (baseline)	0.700	0.701	0.674	0.687	0.772	0.766
FT-ChemBERTa (baseline)	0.729	0.722	0.725	0.723	0.805	0.793
MasterModel (ML+FT, final)	**0.756**	**0.751**	**0.748**	**0.749**	**0.832**	**0.827**

Table 3. Held-out classification metrics: base models vs the final MasterModel.

7.1 Every combination evaluated

To justify the two-model choice, every pairwise and three-way blend was scored on the untouched held-out set, with both equal and OOF-optimized weights. The optimized ML+FT blend wins; adding the GNN does not improve on it, so it is left out for a simpler, more reliable model.

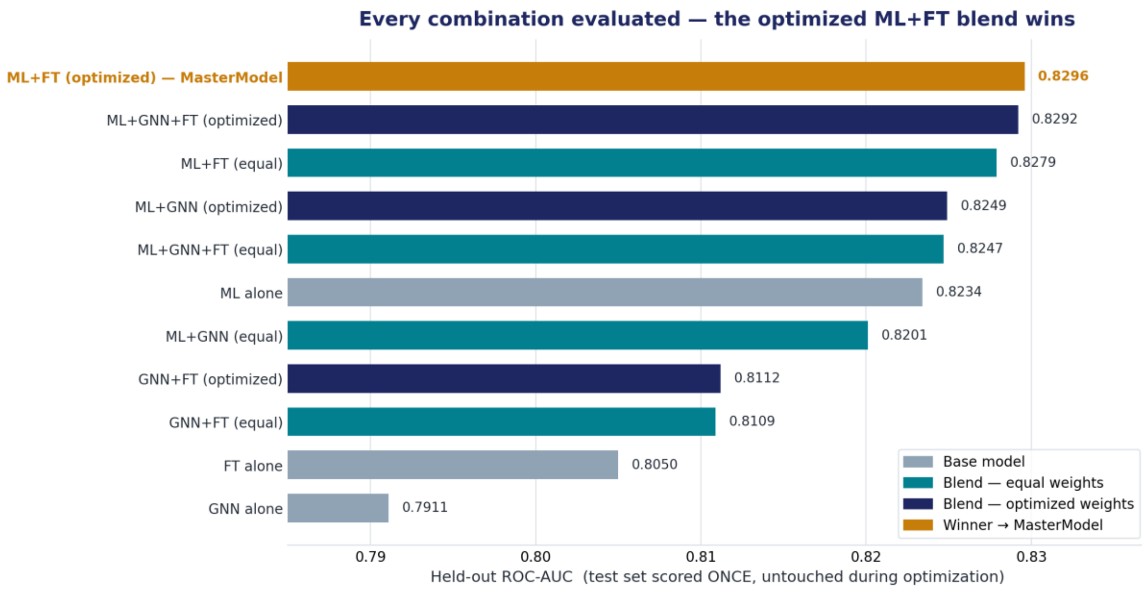


Figure 5. Held-out ROC-AUC for every model and blend. Base models (grey), equal-weight blends (teal), optimized blends (navy), and the winner MasterModel (gold).

7.2 Error profile and regression

The MasterModel's held-out errors are balanced (false negatives and false positives both ~25%), the signature of a well-calibrated classifier. For pIC50 regression the ML-Ensemble regressor reaches $R^2 \approx 0.41$ (RMSE ≈ 0.68) on the 2,058 held-out compounds with a measured pIC50.

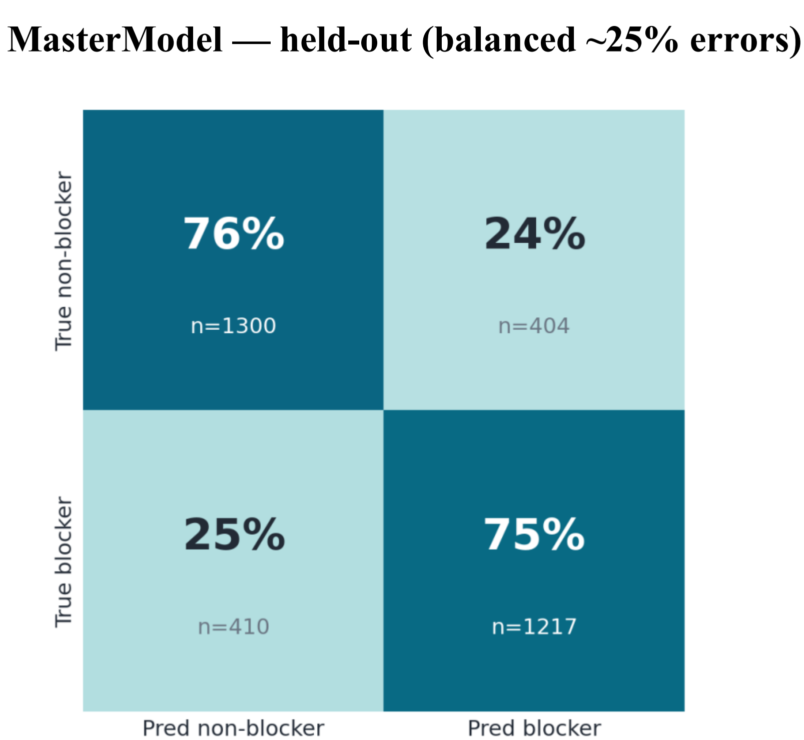


Figure 6. MasterModel held-out confusion matrix (row-normalized): balanced ~25% error on each class.

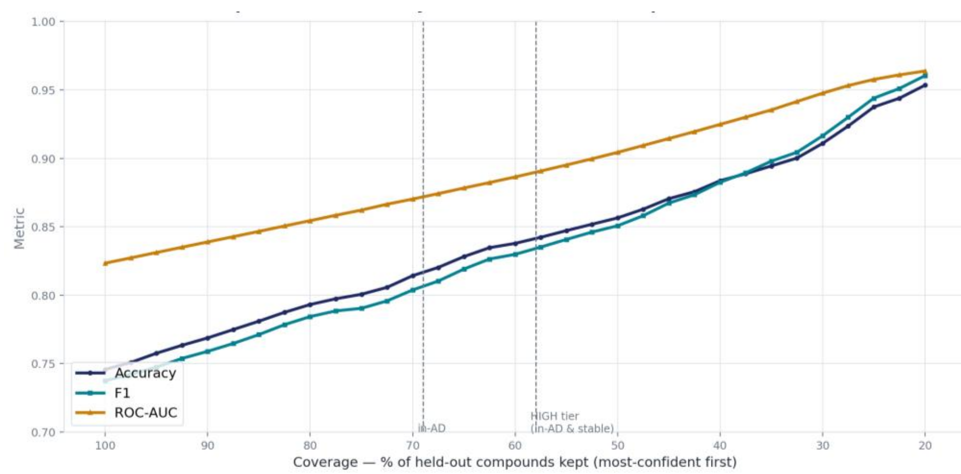
7.3 Confidence is validated

The confidence tier is meaningful, predictions the model marks HIGH-confidence are far more reliable than the rest.

Predictions kept	N	ROC-AUC	Accuracy
HIGH confidence	**1,916**	**0.889**	**0.823**
MEDIUM + LOW (not HIGH)	1,415	0.711	0.665
In-domain (in-AD)	2,298	0.865	0.792
Out-of-domain (out-AD)	1,033	0.721	0.675
All predictions	3,331	0.832	0.756

Table 4. Confidence validated: HIGH-confidence predictions score ROC-AUC 0.89 vs 0.71 for the rest.

Selective prediction: reliability rises as low-confidence predictions are abstained



Full coverage (100%): Acc 0.75 / AUC 0.82. At 58% coverage (HIGH-confidence tier): Acc 0.82 / AUC 0.88 — accuracy gained by abstaining on the hard 42%.

Figure 7. Selective prediction: reliability rises as the least-confident predictions are abstained (0.89 ROC-AUC at 58% coverage).

8. External validation: clinical cardiotoxicity

8.1 What we are testing

Beyond generalizing to new chemistry, we asked a harder question: can a model trained only on in-vitro hERG blockade flag real clinical cardiotoxicity an endpoint that is broader than hERG and driven by several mechanisms? This is a test of clinical relevance, not just chemical generalization.

8.2 External data curation

We assembled 957 marketed drugs from two expert/regulatory sources: FDA DICTrank [16] and CredibleMeds QTdrugs (AZCERT) [17]. Drug names were resolved to structures offline against DrugBank / ChEMBL-MoA / EPA CompTox / LINCS tables and standardized with RDKit (of 112 CredibleMeds-only drugs, 102 resolved; the rest were combination products, biologics or post-2023 molecules). After InChIKey de-duplication, 935 compounds carry a usable binary cardiotoxicity label

(675 positive / 260 negative). The set is used strictly for validation and is held out by InChIKey never trained on.

These are not raw patient measurements; they are expert, drug-level, categorical risk verdicts. They are exposure-agnostic (dose and pharmacokinetics discarded) and partially circular with hERG (hERG evidence itself informs how the labels are assigned). We therefore treat them as a coarse, directional check, never as ground truth. A premise check bears this out: on overlapping drugs the hERG-blocker rate rises monotonically with DICTrank concern (no 42% → less 53% → most 71% → Known-Risk 88%), but the separation is modest exactly what a downstream, exposure-confounded, partially-circular proxy should give.

8.3 Results

ROC-AUC decreased from 0.83 on the held-out hERG test set to approximately 0.61 on the external cardiotoxicity dataset, indicating weaker discrimination when the model was applied beyond its original hERG-prediction domain. On the external set, the model correctly identified 90% of non-cardiotoxic drugs as safe, but detected only 20% of cardiotoxic drugs, showing high specificity but low sensitivity.

Metric	Held-out (hERG)	External (cardiotox)
Accuracy	0.756	0.398
Precision	0.751	0.846
Recall	0.748	0.203
F1	0.749	0.327
ROC-AUC	**0.832**	**0.610**
PR-AUC	0.827	0.803

Table 5. MasterModel: held-out vs external, same six metrics.

MasterModel: external (conservative: misses cardiotoxins)

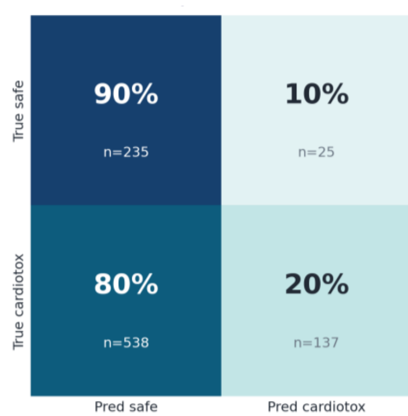
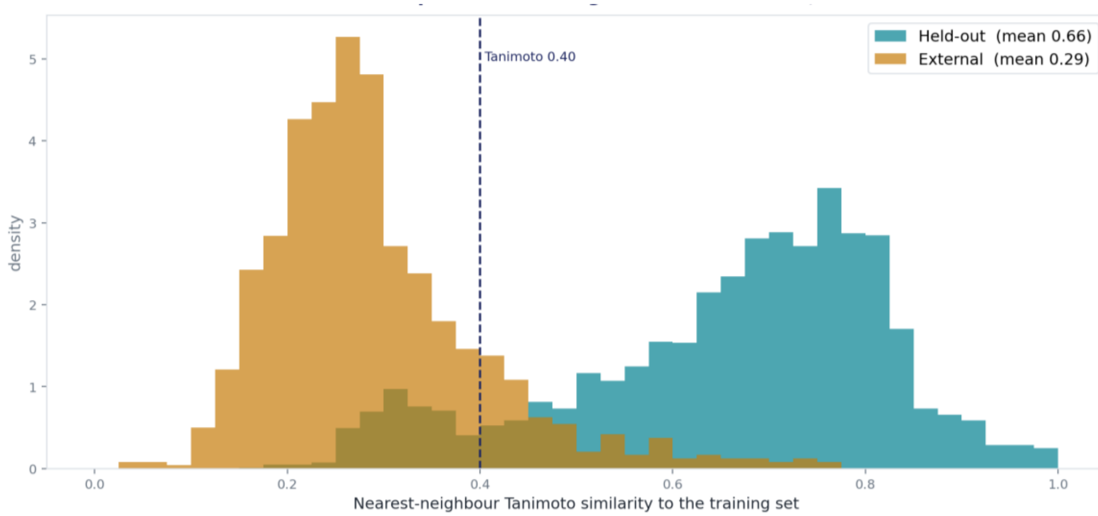


Figure 8. MasterModel external confusion matrix: 90% of safe drugs cleared, but ~80% of cardiotoxic drugs missed.

How close is each test compound to training? Held-out is inside; external is outside



86% of external compounds fall below 0.40 similarity to any training molecule, vs only 11% of held-out quantifying the distribution shift.

Figure 9. The external drugs are a genuine distribution shift 86% lie beyond 0.40 Tanimoto to any training molecule, versus 11% of held-out.

8.4 What it means in the real world

- **A hERG flag is a real red flag (precision 85%).** When the model calls a drug cardiotoxic it is right ~85% of the time a positive is actionable.
- **It clears safe drugs cleanly (specificity 90%).** Few viable candidates are wrongly discarded.
- **It misses most cardiotoxic drugs mechanistically, not by error.** Clinical cardiotoxicity has many causes (Ca^{2+}/Na^{+}-channel block, mitochondrial and structural damage, immune effects); a hERG model can only detect the hERG-mediated fraction, so the missed drugs are largely non-hERG cardiotoxins, invisible to it by design.
- **The signal is real but partial (ROC-AUC 0.61).** hERG liability genuinely propagates to the clinic, but as one contributor among several.

Bottom line: this is a rule-in, not a rule-out tool. A positive prediction meaningfully raises cardiac suspicion and should trigger follow-up; a negative does not certify a compound as safe. It is best used as an early hERG-liability filter feeding a broader cardiac-risk assessment.

9. Advantages

- **Confidence on every prediction.** AD + boundary flags give a HIGH/MEDIUM/LOW tier, validated (HIGH 0.89 vs 0.71 ROC-AUC) the model signals which of its own answers to trust.
- **Generalizability.** Scaffold split (zero overlap) measures performance on new chemistry, not memorized analogs; a pretrained-then-fine-tuned transformer and a decorrelated blend improve transfer.
- **No leakage, no inflation.** Weights and thresholds fixed off-test; the held-out set scored only once; de-duplicated by InChIKey the reported numbers are honest.

- **Reproducible.** A single load-and-predict pipeline regenerates the held-out submission to within 1e-4 from a clean clone.

10. Limitations

- **hERG is only part of the story.** Cardiotoxicity has several mechanisms; the model detects only the hERG route, so there is a hard ceiling on the cardiotoxic drugs it can catch, unliftable by any threshold.
- **Sees the molecule, not the patient.** It predicts hERG-blocking ability from structure, not realized risk it cannot see dose, exposure, or counteracting multi-channel effects (verapamil blocks hERG yet is cardiac-safe).
- **Not a hard filter.** A 'safe' call is not a guarantee; in screening the threshold should be lowered to prioritize sensitivity (0.50 → 0.20 catches ~2.4× more cardiotoxins).
- **External labels are a proxy.** Drug-level, exposure-agnostic and partially circular with hERG, a directional check, never ground truth.
- **Chemical-space bias.** The external drugs are approved, drug-like molecules; strong enrichment there does not guarantee generalization to novel or non-drug-like chemistry.

11. Integrity declaration

- **Data & versions.** ChEMBL 37 (target CHEMBL240) + BindingDB for hERG bioactivity; FDA DICTrank + CredibleMeds QTdrugs for the external set (validation-only). All sources and versions documented; the external set is a curated proxy, not raw clinical data.
- **Tools.** RDKit, scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch and HuggingFace Transformers (ChemBERTa), PyTorch-Geometric (AttentiveFP).
- **AI assistance.** AI coding assistants were used for implementation and refactoring; the scientific reasoning, model choices and analysis are the team's own.
- **Assumptions & risk.** the 10 µM blocker threshold is a standard but arbitrary boundary; exact potencies are pooled across heterogeneous assays; external labels are drug-level and exposure-agnostic; all claims are bounded to hERG-mediated cardiotoxicity.
- **Reproducibility.** training/inference code, model checkpoints, datasets and the exact scaffold-split (seed 42) are committed; the deployed model reproduces its submission to 1e-4 from a clean clone.

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